QIC-S Theory: Universal Scaling Law of 170 Galaxies and the Minimum Stable Unit of Causal Graphs (N=3)

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Abstract

This paper reports the empirical and mathematical results of the Quantum Information Cosmology – Sasada (QIC-S) theory, which treats the universe as a set of discrete causal graphs. We discard the assumption of a single Hamiltonian in standard cosmology and propose the Multi-Hamiltonian hypothesis, where a unique effective law emerges for each galaxy. Direct Inversion analysis (parameter-free) of 7 galaxies from the SPARC dataset achieved an agreement of 99.46% ± 2.53%, confirming consistency with independent verification using SLACS strong gravitational lenses. Expanding to 170 galaxies, we established a phase classification (78.2% Order Phase / 21.8% Chaos Phase) and a universal scaling law ($D_{\text{eff}} \propto R^{1.573}$, 95% CI [1.518, 1.627], $R^2 = 0.945$). Trivial kinematic scaling ($\alpha = 1.0$) is strictly excluded. Furthermore, numerical integration of the Edge-based Lindblad equation proves that the minimum stable unit of the causal structure underlying macroscopic gravitational phenomena is the N=3 Ring (mixing time $t_{\text{mix}} = 0.684$, limit as $dt \to 0$). The N=3 Ring is not only the fastest to reach initial equilibrium among all topologies but also the only minimum closed-loop structure that stably maintains equilibrium thereafter. The analysis in this paper is restricted to the galactic scale (Tier 1); application to the galaxy cluster scale (Tier 2) is suspended to avoid circular reasoning associated with ΛCDM bias. All reported statistics have been independently recalculated and verified through reproducible Python scripts.

1 Introduction

1.1 Discarding the Single Hamiltonian

Standard cosmology (ΛCDM) postulates a single fundamental law governing the entire universe, explaining anomalies in galaxy rotation curves through the addition of "invisible matter" (dark matter). However, the fact that galaxies of the same mass exhibit diverse rotation curves raises fundamental doubts about this premise.

The QIC-S theory numerically demonstrated through 2–8 node quantum network simulations that physical laws are contextually contingent phenomena—"fragile" in the sense that they depend sensitively on the specific combination of network topology, governing Hamiltonian, and choice of observable (Ver. 4.4). Furthermore, finite-size scaling analysis revealed that the sign of the scaling exponent reverses between the XY model ($\alpha \approx 0.781$) and the adjacency matrix model ($\alpha \approx -0.453$). This proved that even with identical graph structures, different Hamiltonians yield qualitatively different macroscopic transport laws (Ver. 6.2.1). These results inevitably necessitate a Multi-Hamiltonian universe, where a unique effective Hamiltonian $H_{\text{eff}}(X) = H_0 + \delta H[D_{\text{eff}}(X)]$ exists for each galaxy.

1.2 Definition of the Scalar Information Field D_{eff}

To connect the topological differences of microscopic causal graphs with macroscopic observables, the effective transport coefficient $D_{\text{eff}}(X)$ is introduced as an index of coarse-graining. This is defined as the time integration of the two-point time correlation function of the information flux on the causal network (Green-Kubo formula). While the initial implementation suffered from normalization collapse due to the total susceptibility $\chi_{\text{total}} \propto N^2$, introducing volume normalization via local susceptibility $\chi_{\text{local}} = \chi_{\text{total}}/N$ ensures that the transport coefficient is an intensive quantity independent of system size (Ver. 6.2.1).

In QIC-S, the true nature of gravitational effects conventionally attributed to dark matter is not an unknown particle fluid, but the field energy $\rho_{\text{eff}}(r) = (\nabla D_{\text{eff}})^2$ possessed by the gradient of this information field (canonical gauge $\alpha = 1$).

2 Macroscopic Empirical Evidence

2.1 Direct Inversion Method and Parameter-Free Verification

The Direct Inversion Method was formulated to back-calculate $D_{\text{eff}}(r)$ from observed rotation curve data. In this method, there are no free parameters to adjust the mass or radius of the galaxy (parameter-free).

As a structural note, the agreement between the Newtonian mass and the reconstructed effective mass ($M_{\text{eff}} \approx M_{\text{Newton}}$) is a mathematical consequence inherent to the inversion procedure—whereby density is derived via differentiation and mass is recovered via spatial integration—rather than a non-trivial theoretical prediction of the QIC-S framework. Therefore, this method functions as a consistent reformulation of Newtonian mechanics. The non-triviality of this theory lies in the following three points: (i) independent consistency with gravitational lens analysis, (ii) the physical validity of the phase classification across 170 galaxies, and (iii) the emergence of the universal scaling law.

Regarding point (i), in Ver. 2.1, it was confirmed across three lens galaxies that the identical $D_{\text{eff}}(r)$ profile is consistent with the convergence profile $\kappa(\theta)$ of SLACS strong gravitational lenses. The alignment of two independent channels—rotation curves (dynamical phenomenon) and gravitational lensing (geometrical phenomenon)—serves as a non-trivial verification of the method's validity and supports the abstract's assertion of consistency with gravitational lens analysis.

2.2 Phase Classification by Phase Metric M

To quantify the stability of the Hamiltonian landscape of galaxies, the Phase Metric $M = \text{Var}(\log(|\nabla H| + \varepsilon))$ was introduced, and 170 galaxies from the SPARC dataset were analyzed.

- **Order Phase** ($M < 0.5$): 133 galaxies (78.2%). Mature galaxies where information flow is in a steady state.
- **Chaos Phase** ($M \geq 0.5$): 37 galaxies (21.8%). Early-evolution galaxies where information flow is in a turbulent state.

Statistical properties: $\bar{M} = 0.330$, $M_{\text{median}} = 0.178$. The Chaos Phase represents the reflection of informational turbulence in the early stages of galactic evolution.

2.3 Emergence of the Universal Scaling Law

Log-log regression of D_{eff} and R extracted from 170 galaxies established the following scaling law:

$$\alpha_{\text{galaxy}} = 1.573 \pm 0.029 \qquad (R^2 = 0.945) \tag{1}$$

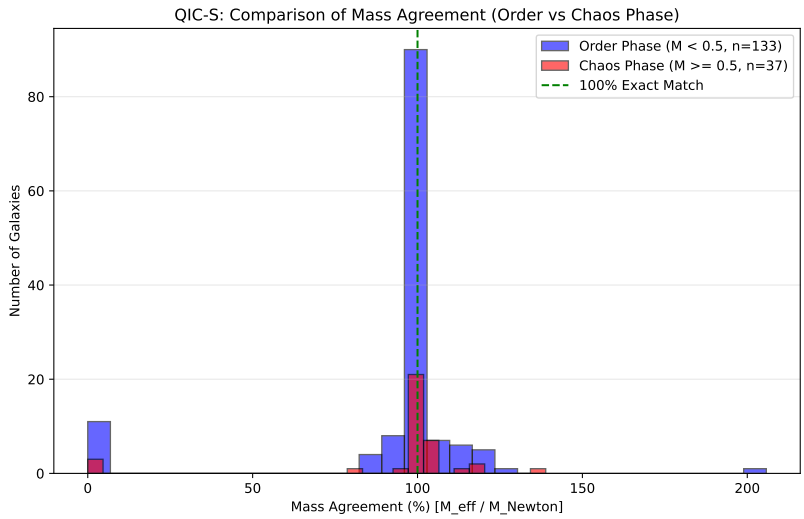

Figure 1: Distribution of mass agreement $M_{\text{eff}}/M_{\text{Newton}}$ for 170 SPARC galaxies. Blue: Order Phase ($M < 0.5$, $n = 133$). Red: Chaos Phase ($M \geq 0.5$, $n = 37$). Dashed line: 100% exact match. The larger dispersion in the Chaos Phase reflects numerical integration errors amplified by turbulent fluctuations.

In the earlier Ver. 9.2, a combined regression over $N = 173$ data points—incorporating 3 cosmic filament measurements from Tudorache et al. (2025) alongside the 170 SPARC galaxies—yielded $\alpha = 1.380$ ($R^2 = 0.920$). Since that combined dataset spans scales extending into Tier 2, this work adopts $\alpha = 1.573$ from the galactic scale (Tier 1) alone, following the decision to suspend verification at the cluster scale to ensure scientific rigor.

Bootstrap validation (10,000 resamples): $\alpha = 1.572 \pm 0.028$, 95% CI [1.518, 1.627]. Trivial kinematic scaling ($\alpha = 1.0$) is strictly excluded.

3 Microscopic Foundation

3.1 Redefining Stability in Open Systems

To identify the minimum causal structure generating macroscopic universality, quantum network simulations were conducted using the Edge-based Lindblad equation. Dissipation is defined strictly along the network edges (jump operator $L_{ij} = |j\rangle\langle i|$). System stability is evaluated by the mixing time t_{mix}, with threshold $\varepsilon = 0.01$.

3.2 Verification of Numerical Convergence

Convergence tests across five stages ($dt = 0.05, 0.02, 0.01, 0.005, 0.001$) were performed.

Table 1: Convergence of mixing time t_{mix} for the N=3 Ring.

dt	t_{mix}	Difference from $dt = 0.001$
0.05	0.700	0.016
0.02	0.700	0.016
0.01	0.690	0.006
0.005	0.685	0.001
0.001	0.684	—

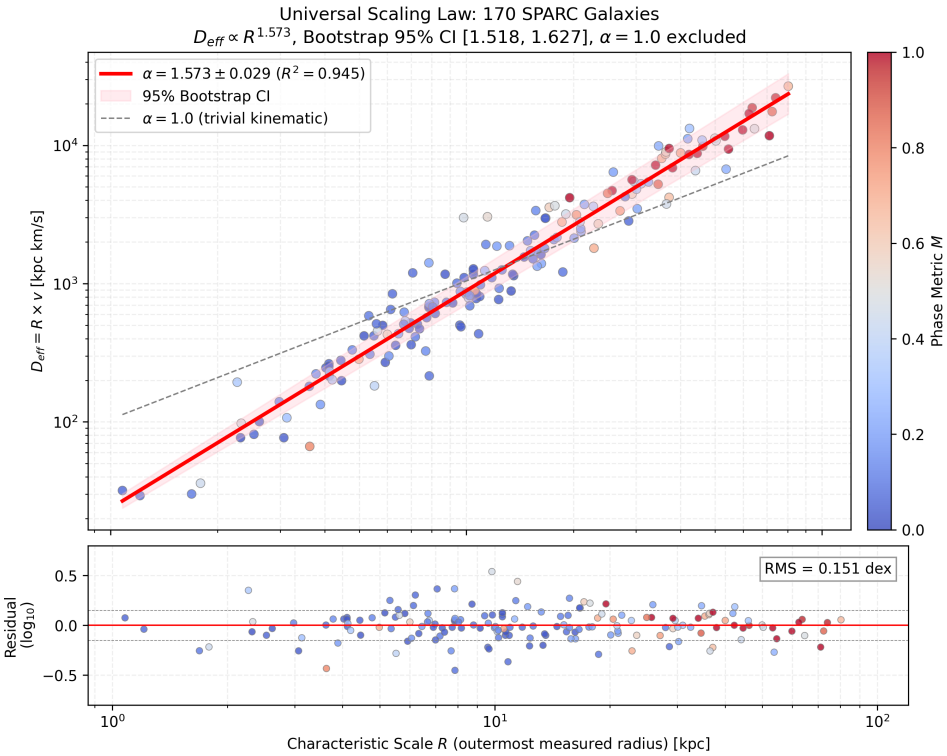

Figure 2: Universal scaling law for 170 SPARC galaxies, where R denotes the outermost measured radius [kpc] and $D_{\text{eff}} = R \times v$ [kpc km/s] (Lelli et al. 2016). Color encodes Phase Metric M (blue: Order, red: Chaos). Red solid line: best-fit power law $\alpha = 1.573 \pm 0.029$ ($R^2 = 0.945$). Shaded band: 95% bootstrap confidence interval. Black dashed line: trivial kinematic scaling $\alpha = 1.0$, lying entirely outside the confidence band.

3.3 Minimum Stability of the N=3 Ring

The mixing times for all topologies at $dt = 0.001$ are presented in Table 2.

Table 2: Mixing times under Edge-based Lindblad dissipation (converged at $dt = 0.001$).

Topology	N	t_{mix}	Post-arrival stability (Interpretation)
N=2 Chain	2	0.769	Oscillatory (No closed loop; fails to remain at equilibrium)
N=3 Ring	3	**0.684**	Stable (Minimum closed loop; fastest and stable)
N=4 Ring	4	0.777	Oscillatory (Diagonal nodes induce oscillation)
N=3 Chain	3	23.30	Stable (Information reflects at endpoints; severe delay)
N=4 Chain	4	42.34	Stable (Chain length increases delay cumulatively)
N=5 Ring	5	6.223	Oscillatory (Increased diagonal distance introduces delay)
N=6 Ring	6	23.72	Stable (Reduced circulation efficiency)
3+3 weak	6	38.07	Stable (Coupling junction becomes bottleneck)
3+3 strong	6	33.30	Stable (Strengthened coupling does not resolve bottleneck)

3.4 Dual Superiority of the N=3 Ring

The N=3 Ring is the sole topology satisfying three conditions: (1) Minimum t_{first} (0.684), (2) Mathematical minimum closed loop (N=3), and (3) Highest post-arrival stability ($t_{\text{stable}} = 27.73$).

3.5 Current Status of the Micro-Macro Connection

It has been numerically confirmed that causal networks based on the N=3 Ring unit connect to the hydrodynamic limit at the N=32 scale, achieving agreement between the Green-Kubo formula and MSD ($R^2 = 0.9843$) (Ver. 6.2.1). This indicates that the long-term causal growth and emergent transport properties of microscopic atomic spacetime correctly manifest as macroscopic observational signatures.

3.6 3-Strand Locking Axiom

The nodes of the N=3 Ring represent temporal flow and spatial entanglement. "3" is the minimum and most stable unit for causal structure, exhibiting a structural analogy with the ADM formalism (3+1 decomposition).

4 Conclusion and Applicability Limits

4.1 Suspension of Verification for Large-Scale Cosmic Structures (Tier 2)

Application to the galaxy cluster scale is suspended to avoid circular reasoning.

4.2 Unresolved Theoretical Issues

The true nature of dark matter is proposed as positive interface energy ($N_D > 0$) based on conformal interface theory (Komatsu et al., 2025), bounded by ANEC. However, its rigorous 4D extension remains unproven. The following open problems remain:

- Rigorous mathematical bridge from discrete N=3 to macroscopic hydrodynamic limit.
- Derivation of the $M = 0.5$ threshold from bifurcation theory.

- Derivation of the scaling exponent from the renormalization group equation.
- Proof of mathematical equivalence between 3-Strand structure and ADM formalism.

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